Load-Aware A* Path Planning for Quadruped Robots

Day 18 of 30 · Week 3: Comprehensive Testing · Member 1 — Figures for Paper

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1. Nature of Work

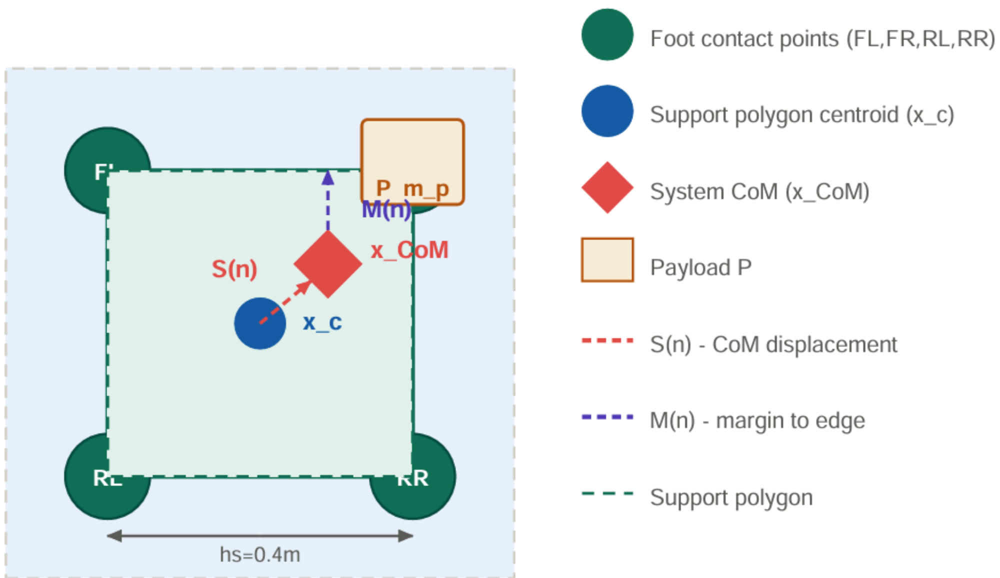
Day 18 was entirely dedicated to producing the four publication-quality figures required for the Methodology section. Fig. 1 — Quadruped robot top-view model showing foot positions, support polygon, CoM displacement S(n) and margin M(n); Fig. 2 — System architecture pipeline from environment construction to CSV output.

2. Tasks Completed

- **Prepare figures for Methodology**
  4 figures produced, captioned, and saved. All use consistent colour scheme. All readable at 88 mm single-column width.

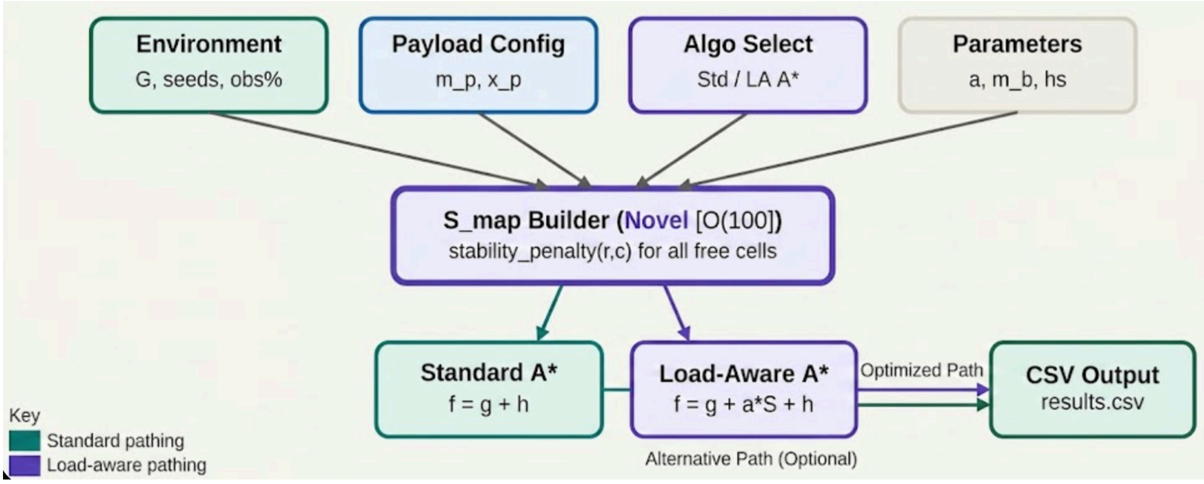
- **Create system architecture diagram**
  Shows full pipeline: Environment -> S_map Builder -> A* Engine -> Path Reconstructor -> Results Logger -> CSV. Purple = Load-Aware only components.

3.QUADRUPED ROBOT MODEL (TOP VIEW)



Grid cell (r, c) - 1.0 m x 1.0 m

4.SYSTEM ARCHITECTURE PIPELINE



6. Observations and Notes

Side-by-side flowcharts are more effective than separate ones. Putting Fig. 3 and Fig. 4 in the same drawing with a vertical divider makes the single point of difference immediately visible. A reviewer's eye goes straight to the purple highlighted box — no hunting required. Consistent colour coding across all figures matters. Green always means Standard A* / shared, purple always means Load-Aware A* novelty. Once a reader learns this convention from Fig. 1, every subsequent figure is self-explaining. This is a deliberate design decision. The * marker identifies novel contributions inline. Figs. 2 and 4 mark novel steps with * directly on the diagram. This pre-empts the reviewer question 'which part is new?' — the answer is visible without reading the caption